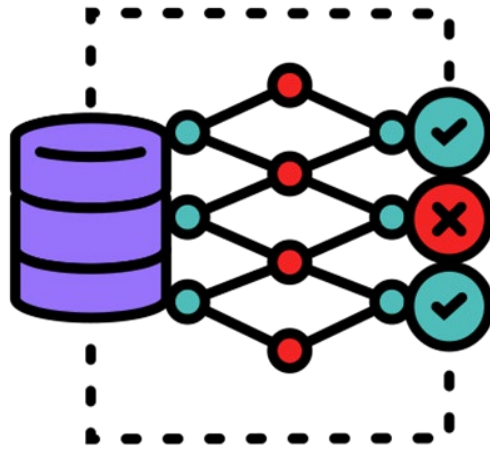




DECISION SUPPORT SYSTEMS

2025

FINAL BANK

(Solved)




SENIOR
25



FINAL MCQ BANK

Choose the correct answer:

1.	an interrelated set of components including people, activities, technology, and procedures that are designed or intended to achieve a predefined purpose			
	a) system	b) Goal	c) Resources	d) None
2.	is a process by which organizational goals are achieved through the use of resources.			
	a) system	b) Management	c) Resources	d) Goal
3.	Resources refer to.....			
	a) Inputs	b) Output	c) Outputs / Inputs	d) None
4.	Goal Attainment refer to.....			
	a) Inputs	b) Output	c) Outputs / Inputs	d) None
5.	Measuring Success refer to			
	a) Inputs	b) Output	c) Outputs / Inputs	d) None
6.	 awareness or familiarity gained by experience of a fact or situation .			
	a) Knowledge	b) Information	c) Data	d) None
7.	 comprises things that are held to be true in each context and that drive us to action if there were no impediments			
	a) Knowledge	b) Information	c) Data	d) None
8.	 defined as processed data is an increment in knowledge : it contributes to the general framework of concepts and facts that we know.			
	a) Knowledge	b) Information	c) Data	d) None
9.	 are only the raw facts, the material for obtaining information . <u>Information systems use data stored in computer databases to provide needed information.</u>			
	a) Knowledge	b) Information	c) Data	d) None
10.	 an organized collection of interrelated data reflecting a major aspect of a firm's activities			
	a) Database	b) DBMS	c) Downsizing	d) None
11.	 is an elementary activity conducted during business Operations.			
	a) Transaction	b) system	c) Modelling	d) None
12.	TPS may work either inmode, processing accumulated transactions at a single time later on or inmode, processing incoming transactions immediately.			
	a) batch, on-line	b) on-line, batch	c) system, Modelling	d) None
13.	TPS today generally work inmode by immediately processing a firm's business transactions.			
	a) on-line	b) batch	c) Transaction	d) None



14.	 The objective of management reporting systems is to provide routine information to managers.							
	a)	Transaction processing systems (TPS)	b)	Management Reporting Systems (MRS)	c)	Decision Support Systems (DSS)	d)	Executive Information Systems (EIS)
15.	 Managers receive performance reports within their specific areas of responsibility							
	a)	Transaction processing systems (TPS)	b)	Management Reporting Systems (MRS)	c)	Decision Support Systems (DSS)	d)	Executive Information Systems (EIS)
16.	 Generally, these reports provide internal information rather than spanning corporate boundaries.							
	a)	Transaction processing systems (TPS)	b)	Management Reporting Systems (MRS)	c)	Decision Support Systems (DSS)	d)	Executive Information Systems (EIS)
17.	 They report on the past and the present, rather than projecting the future							
	a)	Transaction processing systems (TPS)	b)	Management Reporting Systems (MRS)	c)	Decision Support Systems (DSS)	d)	Executive Information Systems (EIS)
18.	 These systems facilitate a dialog between the user , who is considering alternative problem solutions , and the system that provides built-in models and access to databases .							
	a)	Transaction processing systems (TPS)	b)	Management Reporting Systems (MRS)	c)	Decision Support Systems (DSS)	d)	Executive Information Systems (EIS)
19.	Thedatabases are often extracted from the general databases of the enterprise or from external databases .							
	a)	Transaction processing systems (TPS)	b)	Management Reporting Systems (MRS)	c)	Decision Support Systems (DSS)	d)	Executive Information Systems (EIS)
20.	 support top managers with conveniently displayed summarized information, customized for them.							
	a)	Transaction processing systems (TPS)	b)	Management Reporting Systems (MRS)	c)	Decision Support Systems (DSS)	d)	Executive Information Systems (EIS)
21.	Executive Information Systems are use to							
	a)	Monitor the performance of the organization.	b)	Assess the business environment.	c)	Develop strategic directions for the company's future.	d)	All the above
22.	is to facilitate communication between the members of an organization and between the organization and its environment							
	a)	Transaction processing systems (TPS)	b)	Management Reporting Systems (MRS)	c)	Decision Support Systems (DSS)	d)	Office Information Systems (OIS)
23.	 are system that employs knowledge about its application domain and uses an inferencing (reason) procedure to solve problems that would otherwise require human competence or expertise .							
	a)	Transaction processing systems (TPS)	b)	Management Reporting Systems (MRS)	c)	Decision Support Systems (DSS)	d)	Expert systems (ES)



24.	_____ is a choice made between alternative courses of action in a situation of uncertainty.			
	a) Decision	b) Decision making	c) Support	d) DSS
25.	 Process of choosing amongst alternative courses of action for the purpose of attaining a goal or goals.			
	a) Decision Making	b) Decision	c) Intelligence	d) None
26.	 is a cognitive process that results in the selection of a course of action among several alternative scenarios.			
	a) Decision Making	b) Decision	c) Intelligence	d) None
27.	A decision problem is characterized by.....			
	a) Decision alternatives	b) States of nature	c) Payoffs	d) All of them
28.	It contains data from various sources, including internal data from the organization, the data generated by different applications, and the external data mined from the Internet			
	a) DSS Database	b) DSS Software System	c) DSS User Interface	d) None
29.	It consists of various mathematical and analytical that are used to analyze complex data, thereby producing the required information.			
	a) DSS Database (DSS DB)	b) DSS Software System (DSS SW)	c) DSS User Interface (DSS UI)	d) DSS Dialogue
30.	 It is an interactive graphical interface which makes the interaction easier between the DSS and its users. It displays the results (output) of the analysis in various forms, such as text, table, charts or graphics			
	a) DSS Database	b) DSS Software System	c) DSS User Interface (DSS UI)	d) None
31.	 Supports decision making by analyzing given time-series of data and returning new information gained by those analysis. or <u>It is used to query a database or data warehouse to seek specific answers for specific purposes.</u>			
	a) data-driven DSS	b) document-driven DSS	c) Knowledge-driven DSS	d) Model- driven DSS
32.	The purpose of such a DSS is to search web pages and find documents on a specific set of keywords or search terms. Or <u>the usual technology used to set up such DSSs are via the web or a client/server system</u>			
	a) data-driven DSS	b) document-driven DSS	c) Knowledge-driven DSS	d) Model- driven DSS
33.	 The typical deployment technology used to set up such systems could be client/server systems, the web, or software running on stand-alone PCs. Or <u>It is essentially used to provide management advice or to choose products/services.</u>			
	a) data-driven DSS	b) document-driven DSS	c) Knowledge-driven DSS	d) Model- driven DSS
34.	 These are used by managers and staff members of a business, or people who interact with the organization, for several purposes depending on how the model is set up - scheduling, decision analyses			
	a) data-driven DSS	b) document-driven DSS	c) Knowledge-driven DSS	d) Model-driven DSS
35.	 is a technology-driven process for analyzing data and delivering actionable information that helps executives, managers and workers make informed business decisions.			
	a) BI	b) DSS	c) ERD	d) None



36.	 Repetitive and routine , and decision makers can follow a definite procedure for handling them to be efficient							
	a)	Structured (Operational)	b)	Semi-Structured (Tactical)	c)	Unstructured (Strategic)	d)	None
37.	 Only part of the problem has a clear-cut answer provided by an accepted procedure.							
	a)	Structured (Operational)	b)	Semi-Structured (Tactical)	c)	Unstructured (Strategic)	d)	None
38.	Each of these decisions is novel, important, and nonroutine , and there is no well-understood or agreed-on procedure for making them							
	a)	Structured (Operational)	b)	Semi-Structured (Tactical)	c)	Unstructured (Strategic)	d)	None
39.	Decisions are those in which the decision maker must provide judgment , evaluation, and insights into the problem definition.							
	a)	Structured (Operational)	b)	Semi-Structured (Tactical)	c)	Unstructured (Strategic)	d)	None
40.	It is assumed that complete knowledge is available so that the decision-maker knows exactly what the outcome of each course of action will be (as in a deterministic environment) example for decision making Under.....							
	a)	certainty	b)	risk	c)	uncertainty	d)	None
41.	The decision-maker is viewed as a perfect predictor of the future because it is assumed that there is only <i>one outcome</i> for each alternative example for decision making Under							
	a)	certainty	b)	risk	c)	uncertainty	d)	None
42.	Every time you park downtown, you get a parking ticket because you exceed the time limit on the parking meter <i>although once it did not happen</i> , this is an example for decision making under (بتروح تركز بعريبتك ولو قعدت اكثر من الحده المتاحه بتاخذ غرامه. وكل مره بتركز هناك وتعدي مدتک المتاحه بس في مره رکنت ربع ساعة بس وبرضه خدت غرامه عشان هو متأكد انه انت اللي كل مره بتركز كثير)							
	a)	certainty	b)	risk	c)	uncertainty	d)	None
43.	Decision-making under The decision maker does not know, or cannot estimate, the probability of occurrence of the possible outcomes example for decision making under.....							
	a)	certainty	b)	risk	c)	uncertainty	d)	compress
44.	The decision-maker must consider several possible outcomes for each alternative, each with a given probability of occurrence under.....							
	a)	certainty	b)	risk	c)	uncertainty	d)	None
45.	The output of Traditional DSS are decisions based on data							
	a)	Quantitative	b)	Qualitative	c)	Both A&B	d)	None
46.	The output of Knowledge-based DSS are decisions based on..... data							
	a)	Quantitative	b)	Qualitative	c)	Both A&B	d)	None
47.	phase of DSS process consists of choosing among solution alternatives.							
	a)	Intelligence	b)	Design	c)	Choice	d)	Implementation
48.	Sensitivity analysis is applied in which phase of DSS process?							
	a)	Intelligence	b)	Design	c)	Choice	d)	Implementation



49.	 phase of DSS process consists of discovering, identifying, and understanding the <u>problems occurring</u> in the organization.			
	a) Intelligence	b) Design	c) Choice	d) Implementation
50.	 phase of DSS process involves identifying and exploring various <u>solutions</u> to the problem			
	a) Intelligence	b) Design	c) Choice	d) Implementation
51.	 involves identifying and exploring various <u>solutions</u> to the problem.			
	a) Intelligence	b) Design	c) Choice	d) Implementation
52.	 phase of DSS process involves making the chosen alternative work and continuing to monitor how well the solution is working			
	a) Intelligence	b) Design	c) Choice	d) Implementation
53.	Model are used to provide answers to what-if situations occurring frequently in an organization			
	a) Sensitivity Analysis	b) Statistical	c) Forecasting	d) Backward Analysis
54.	 Model are known as goal seeking analysis, the technique followed in these models is just opposite to the technique applied in sensitivity analysis models.			
	a) Sensitivity Analysis	b) Statistical	c) Forecasting	d) Backward Analysis
55.	Model are contain a wide range of statistical functions, such as mean, median, mode, deviations etc. <u>These models are used to establish, relationships between the occurrences of an event and various factors related to that event</u>			
	a) Sensitivity Analysis	b) Statistical	c) Forecasting	d) Backward Analysis
56.	 Model are used to find optimum value for a target variable under given circumstances.			
	a) Sensitivity Analysis	b) Optimization Analysis	c) Forecasting	d) Backward Analysis
57.	 Model are use various forecasting tools and techniques.			
	a) Sensitivity Analysis	b) Optimization Analysis	c) Forecasting	d) Backward Analysis
58.	may work either in batch mode, processing accumulated transactions at a single time later on, or in on-line mode, processing incoming transactions immediately.			
	a) TPS	b) MIS	c) DSS	d) EIS
59.	support top managers with conveniently displayed summarized information, customized for them.			
	a) TPS	b) MIS	c) DSS	d) EIS
60.	The main objective of..... is to facilitate communication between the members of an <u>organization</u> and between the organization and its environment.			
	a) TPS	b) MIS	c) DSS	d) OIS
61.	 These systems facilitate a dialog between the user , who is considering <i>alternative problem solutions</i> , and the system that provides built-in models and access to databases.			
	a) TPS	b) MIS	c) DSS	d) EIS
62.	is the increase in the expected profit that would result if one knew with certainty which state of nature would occur			
	a) AHP	b) EV	c) ROMC	d) EVPI



63.	If probabilistic information regarding the states of nature is available, one may use the.....							
	a)	AHP	b)	EV	c)	ROMC	d)	EVPI
64.	 is done by comparing model's output and the actual behavior of the event .							
	a)	Model validation	b)	Modeling	c)	Assumptions	d)	Forecasts
65.	 is the process of identifying an appropriate model for a prospective model-driven decision support system, Or <u>once _____ is done, it's vital to validate the selected model, to ensure it works well and generates appropriate results.</u>							
	a)	Model validation	b)	Modeling	c)	Assumptions	d)	Forecasts
66.	Analysis doesn't take into consideration the long-term response of a system. Or <u>It takes a single snapshot of a situation</u> and assumes that it will remain stable all through and won't change.							
	a)	Static	b)	Dynamic	c)	Explanatory	d)	none
67.	 Analysis is testing a program or a software system in real-time . Or <u>This method considers that the situation changes over time, due to any reason, such as cost, rules and regulations, time, etc.</u>							
	a)	Static	b)	Dynamic	c)	Explanatory	d)	none
68.	Model Describes and explains why something is the way it is and why and how it works.							
	a)	Descriptive (Explanatory)	b)	Contemplative	c)	Algebraic	d)	none
69.	 Model Forecasts results or outcomes that may be produced from a specific set of parameters.							
	a)	Descriptive (Explanatory)	b)	Contemplative	c)	Algebraic	d)	none
70.	 Model high-level modeling system for solving complex equations . It is employed to optimize a variable or equation. The best part is that it can handle several simultaneous equations.							
	a)	Descriptive (Explanatory)	b)	Contemplative	c)	Algebraic	d)	none
71.	 summarize the anticipated financial results for a specific period in the future.							
	a)	Pro Forma	b)	Break-Even	c)	AHP model	d)	Ratio
72.	 a financial tool that helps businesses determine the <u>sales volume</u> or revenue needed to cover all costs associated with producing and selling a product or service.							
	a)	Pro Forma	b)	Break-Even	c)	AHP model	d)	Ratio
73.	a multi-criteria decision technique (MCDM) that combines quantitative and qualitative factors when evaluating alternatives.							
	a)	Analytical Hierarchy Process (AHP)	b)	Decision Tree (DT)	c)	Multi-attribute utility analysis (MAUA)	d)	Payoff Table (PT)
74.	 emphasizes the quantification of individual preferences using utility functions .							
	a)	Analytical Hierarchy Process (AHP)	b)	Decision Tree (DT)	c)	Multi-attribute utility analysis (MAUA)	d)	Payoff Table (PT)



75.	The consequence resulting from a specific <i>combination of a <u>decision alternative and a state of nature</u></i> is a					
	a)	Analytical Hierarchy Process (AHP)	b)	Decision Tree (DT)	c)	Multi-attribute utility analysis (MAUA)
					d)	Payoff
76.	A is a graphical representation of the decision problem.					
	a)	Analytical Hierarchy Process (AHP)	b)	Decision Tree (DT)	c)	Multi-attribute utility analysis (MAUA)
					d)	Payoff Table (PT)
77.	Each decision tree has two types of nodes; round nodes correspond to while square nodes correspond to					
	a)	Decision alternatives, States of nature	b)	States of nature, Decision alternatives	c)	Utility functions, multi-criteria
					d)	Multi-criteria, Utility functions
78.	In profit payoff, Choose decision with the maximum of the maximum payoffs.					
	a)	Maximax (optimistic)	b)	Maximin (pessimistic)	c)	Minimax (regret)
					d)	None
79.	In profit payoff, Choose decision with the maximum of the minimum payoffs.					
	a)	Maximax (optimistic)	b)	Maximin (pessimistic)	c)	Minimax (regret)
					d)	None
80.	In profit payoff, Choose decision with the minimum of the maximum regrets for each alternative.					
	a)	Maximax (optimistic)	b)	Maximin (pessimistic)	c)	Minimax (regret)
					d)	None
81.	MAUA is considered.....tool					
	a)	Financial	b)	budget analysis	c)	decision analysis
					d)	optimization
82.	 <u>simple</u> forecasting that provides limited accuracy					
	a)	Naïve Exploration	b)	Judgment Methods	c)	Regression
					d)	Exponential Smoothing
83.	 the perceptions and opinions of experts instead on hard data, long term forecasting.					
	a)	Naïve Exploration	b)	Judgment Methods	c)	Regression
					d)	Exponential Smoothing
84.	 the predictions are based upon the historical values .					
	a)	Naïve Exploration	b)	Judgment Methods	c)	Moving Average
					d)	Exponential Smoothing
85.	alters the historical data mathematically to better reflect the assumptions of a decision maker					
	a)	Naïve Exploration	b)	Judgment Methods	c)	Moving Average
					d)	Exponential Smoothing
86.	 considers the economic variables that are measured at consecutive intervals of time . Or <u>It is believed that the knowledge of past behavior of the variable at successive intervals of time will help understand the behavior of the variables in future better.</u>					
	a)	Naïve Exploration	b)	Judgment Methods	c)	Moving Average
					d)	Time Series Extrapolation



87.	 use of linear and multiple regressions to establish <i>cause and effect</i> relationships							
	a)	Naïve Exploration	b)	Judgment Methods	c)	Regression & Econometric Models	d)	Exponential Smoothing
88.	 conduct experiments to identify conditions or situations that approximate the actual conditions.							
	a)	simulation model	b)	Network and optimization models	c)	Forecasting model	d)	none
89.	 All models are made up of three basic components : decision <u>variables</u> , uncontrollable <u>variables</u> (and/or parameters), and result (outcome) <u>variables</u> .							
	a)	Mathematical models	b)	Network and optimization models	c)	Forecasting model	d)	none
90.	is a model-driven DSS to help a person decide how much he or she needs to <u>invest</u> each month <u>for retirement</u> .							
	a)	Fidelity	b)	Microsoft Carpoint	c)	Stockfinder	d)	TCB Works
91.	 a web-based system that allows users to make pair-wise comparisons of <u>car</u> models.							
	a)	Fidelity	b)	Microsoft Carpoint	c)	Stockfinder	d)	TCB Works
92.	a knowledge-driven web-based decision support systems, offering 25 different <u>decision guides</u> on <u>diverse topics</u> , including choosing pets, bikes and business schools.							
	a)	Fidelity	b)	Microsoft Carpoint	c)	Netscape Decision Guides	d)	TCB Works
93.	 It's a data-driven DSS to help investors identify stocks based on various criteria, including industry type, price and earnings							
	a)	Fidelity	b)	Microsoft Carpoint	c)	Stockfinder	d)	TCB Works
94.	 A communication-driven web-based DSS, it enables people to interact, discuss and make both structured and multi-criteria decisions							
	a)	Fidelity	b)	Microsoft Carpoint	c)	Stockfinder	d)	TCB Works
95.	 is a collection of related data and <u>data is a collection of facts and figures</u> that can be processed to produce information							
	a)	Database	b)	DBMS	c)	Downsizing	d)	None
96.	 stores data in such a way that it becomes <u>easier to retrieve</u> , manipulate, and produce information.							
	a)	Database	b)	DBMS	c)	Downsizing	d)	None
97.	 is an organized approach to design specialized decision support systems, more precisely their <u>user interfaces</u> . Or <u>is a systematic move</u> towards designing the representation, operation, memory and control mechanisms of a large decision support system.							
	a)	MAUA	b)	AHP	c)	GUI	d)	ROMC
98.	 is about presenting information or results for that matter, in a structured way. Or <u>provides a base to the decision makers backed by concrete information</u> , helping them interpret DSS outputs. And <u>can be in the form of a table, graph, map, chart or a text document</u> and each value on a map or a table communicates decision making context.							
	a)	Representation	b)	operation	c)	memory aid	d)	none
99.	is stage in user interface design focuses on specific tasks performed by/with a decision support system. Or <u>an involve one activity or many, depending upon the specific needs</u> of the decision maker.							
	a)	Representation	b)	operation	c)	memory aid	d)	none



100.	 measures the extent to which the system meets its objectives .			
	a) reliability	b) effectiveness	c) efficiency	d) completeness
101.	DSS support all organization levels except level.			
	a) strategic	b) operational	c) management	d) tactical
102.	The decision-making process can be defined as the activity that leads to the solution of a decision-making problem involving at least Alternatives			
	a) one	b) two	c) three	d) four
103.	Data availability is more in which of the following?			
	a) strategic	b) operational	c) management	d) tactical
104.	 improving communication and coordination among people with organizations to improve organization performance and solve problems.			
	a) web-based DSS	b) GDSS	c) KBDSS	d) ODSS
105.	The output of the DSS <u>design</u> phase is			
	a) alternatives	b) Problem statement	c) criteria	d) solutions
106.	If a company has fixed costs of \$100,000, variable costs of \$10 per unit, and a selling price of \$20 per unit, This means that the company must sell units to cover all of its costs. $\text{Break-even} = \frac{\text{Fixed cost}}{\text{selling price} - \text{variable cost}} = \frac{100000}{20 - 10}$			
	a) 1000	b) 2000	c) 5000	d) 10000
107.	If a company has fixed costs of \$100,000, variable costs of \$20 per unit, and a selling price of \$2000 per 50 units , this means that the company must sell units to cover all its costs (BEA). لازم اخلي ال selling price تبقى per unit بس (\$2000/50 = 40 per unit) $\text{Break-even} = \frac{\text{Fixed cost}}{\text{selling price} - \text{variable cost}} = \frac{100000}{40 - 20}$			
	a) 1000	b) 2000	c) 5000	d) 10000
108.	If a company has fixed costs of \$100,000, variable costs of \$25 per unit, and a selling price of \$2000 per 40 units , this means that the company must sell units to cover all its costs (BEA). لازم اخلي ال selling price تبقى per unit بس (\$2000/40 = 50 per unit) $\text{Break-even} = \frac{\text{Fixed cost}}{\text{selling price} - \text{variable cost}} = \frac{100000}{50 - 25}$			
	a) 1000	b) 4000	c) 5000	d) 10000
109.	 analysis can help you to compare your company's performance to that of other companies in your industry .			
	a) Pro Forma	b) Break-Even	c) AHP model	d) Ratio
110.	is an umbrella term that combines architectures, tools, databases, analytical tools, applications, and methodologies.			
	a) DSS	b) BI	c) Data mining	d) MSS



111.	The process of is based on the transformation of data to information, then to decisions, and finally to actions.			
	a) DSS	b) BI	c) Data mining	d) MSS
112.	Managers and specialists use the <u>concept of a system</u> frequently and yet it is hard for most of us to define and understand the concept.			
	a) EIS	b) TPS	c) MIS	d) DSS
113.	Systems also have mechanisms to provide a means of controlling the operation of the system.			
	a) input	b) output	c) feedback	d) results
114.	Depending on the type of, the information output may take the form a <u>query response, decision outcome, expert-system advice, transaction document, or a report.</u>			
	a) DSS	b) MIS	c) IS	d) ES
115.	The objectives of a system are realized in its			
	a) input	b) output	c) feedback	d) results
116.	 measuring performance against the planned objectives and initiating corrective action , if needed			
	a) planning	b) leadership	c) controlling	d) organizing
117.	 establishing goals and selecting the actions needed to achieve them over a specific period of time			
	a) planning	b) leadership	c) controlling	d) organizing
118.	process of establishing worker relationships allows workers to work together to achieve their organizational goals.			
	a) planning	b) leadership	c) controlling	d) organizing
119.	including the people in the organization to contribute to its goals			
	a) planning	b) leadership	c) controlling	d) organizing
120.	 Recruiting and selecting employees for positions within the company (within teams and departments)			
	a) planning	b) leadership	c) Staffing	d) organizing
121.	clarifies the computer's role in aiding rather than replacing the decisionmaker.			
	a) Decision	b) support	c) systems	d) DSS
122.	Its purpose is to help conduct a meeting , or for users to collaborate driven DSS.			
	a) Office	b) Data	c) Communication	d) document
123.	is an approach to discovery and problem solving by employing practical methods.			
	a) Heuristics	b) Inference engine	c) Knowledge acquisition	d) Case reasoning
124.	 used to determine the <u>distance between or relationship among various components.</u>			
	a) Heuristics	b) Inference engine	c) Knowledge acquisition	d) Case reasoning
125.	“What techniques will it use?” used in for designing web-based DSS.			
	a) conceptualization	b) Problem identification	c) Feasibility analysis	d) SW Development



126.	 means that a decision support system must be resourceful enough to perform the entire range of tasks that a decision maker needs to perform when making a decision using DSS.			
	a) Adaptability	b) Versatility	c) Quality	d) Fatigue
127.	 helps you to identify <u>all possible causes</u> of the problem.			
	a) Cause-and-Effect diagram	b) Pareto Chart	c) Decision tree	d) None
128.	 helps you to prioritize and identify the causes <u>with the highest effect</u> .			
	a) Cause-and-Effect diagram	b) Pareto Chart	c) Decision tree	d) None
129.	The phase involves defining the purpose and objectives of the DSS and identify the specific decision-making problems or opportunities that the DSS will address			
	a) Initialization	b) Decision Oriented Diagonal	c) Feasibility study	d) None
130.	The to knowledge management attempts to codify organizational knowledge through formalized controls, processes, and technologies			
	a) process approach	b) Practice approach	c) Hybrid approach	d) None
131.	Organizations that adopt the may implement explicit policies governing how knowledge is to be collected, stored, and disseminated throughout the organization			
	a) process approach	b) Practice approach	c) Hybrid approach	d) None
132.	 the knowledge may be typically static in nature			
	a) process approach	b) Practice approach	c) Hybrid approach	d) None
133.	The to knowledge management assumes that a great deal of organizational knowledge is tacit in nature and that formal controls, processes, and technologies are not suitable for transmitting this type of understanding			
	a) process approach	b) Practice approach	c) Hybrid approach	d) None
134.	The focus of approach is to build the social environments or communities of practice necessary to facilitate the sharing of tacit understanding			
	a) process approach	b) Practice approach	c) Hybrid approach	d) None
135.	 is typically adopted by companies that provide highly customized solutions to unique problems.			
	a) process approach	b) Practice approach	c) Hybrid approach	d) None
136.	Many organizations use a			
	a) process approach	b) Practice approach	c) Hybrid approach	d) None
137.	 is any combination of hardware and software that enhances groupwork either in direct or indirect support of decision making			
	a) GDSS (group decision support system)	b) GSS (group support system)	c) DSS (decision support system)	d) DGSS
138.	 an interactive computer-based system that facilitates the solution of semi structured or unstructured problems by a group of decision makers .			
	a) GDSS (group decision support system)	b) GSS (group support system)	c) DSS (decision support system)	d) DGSS



139.	_____ is the systematic and active management of ideas, information, and knowledge residing in an organization's employees							
a)	Knowledge management (KM)	b)	knowledge process (KP)	c)	Decision Support System (DSS)	d)	All are true	
140.	_____ process that helps organizations <u>identify, select, organize, disseminate, and transfer</u> important information and expertise that are part of the organization's memory and that typically reside within the organization in an unstructured manner.							
a)	KM (knowledge management)	b)	KP (knowledge process)	c)	DSS (Decision Support System)	d)	All are true	
141.	_____ is a <u>family of methods</u> that attempt to allow machines to acquire knowledge for problem-solving by showing them historical cases							
a)	Machine learning	b)	supervised learning	c)	Unsupervised Learning	d)	Reinforcement Learning	
142.	_____ is used for making predictions from data . To be able to do that, we need to know what to predict, which is also known as the target variable							
a)	Machine learning	b)	supervised learning	c)	Unsupervised Learning	d)	Reinforcement Learning	
143.	_____ is the science of decision making. It is about learning the optimal behaviour in an environment to obtain maximum reward .							
a)	Machine learning	b)	supervised learning	c)	Unsupervised Learning	d)	Reinforcement Learning	
144.	_____ is self-organized learning. Its main aim is to explore the <u>underlying</u> patterns and predicts the output							
a)	Machine learning	b)	supervised learning	c)	Unsupervised Learning	d)	Reinforcement Learning	
145.	_____ is a problem-solving methodology that attempts to mimic how our <u>brains</u> function.							
a)	Machine learning	b)	Neural computing	c)	Unsupervised Learning	d)	Reinforcement Learning	
146.	The human brain is composed of special cells called _____							
a)	Neurons	b)	Axon hillock	c)	Cell body	d)	Axon	
147.	In Biological and artificial neural, Dendrites = _____, and Axon = _____							
a)	Input/Input	b)	Input/Output	c)	Output/input	d)	output/ output	
148.	In an artificial neuron, Synapse = _____							
a)	Node	b)	Input	c)	Output	d)	Weights/ interconnections	
149.	In an artificial neuron, Cell Nucleus = _____							
a)	Node	b)	Input	c)	Output	d)	Weights/ interconnections	
150.	In Artificial neural networks, The basic processing unit is the _____							
a)	Neurons	b)	Axon hillock	c)	Cell body	d)	Axon	
151.	A _____ is composed of processing elements organized in different ways to form the network's structure.							
a)	Neural network	b)	Neural computing	c)	Artificial Neurons	d)	None	



152.	The processing elements (PEs) of an ANN are _____			
	a) Neural network	b) Neural computing	c) Artificial Neurons	d) None
153.	Decision at the top level tend to be more _____			
	a) Structured	b) Semi-structured	c) Unstructured	d) None
154.	Each ANN is composed of a collection of neurons grouped in three layers _____.			
	a) input	b) Intermediate	c) Output	d) All
155.	Each _____ corresponds to a single attribute			
	a) input	b) Intermediate	c) Output	d) All
156.	The _____ of the network contain the solution to a problem			
	a) input	b) Intermediate	c) Output	d) All
157.	The resulting model from neural computing is often called _____.			
	a) Neural network	b) Artificial Neural Network (ANN)	c) Neural computing	d) A & B
158.	_____ A population of potential solutions (individuals or chromosomes) is <i>randomly generated</i> . Each solution represents a possible solution to the problem at hand.			
	a) Initialization	b) Evaluation	c) Selection	d) Crossover
159.	_____ Each individual in the population is evaluated for its fitness, which is a measure of how well it solves the problem. The fitness function is problem specific and reflects the objective or goal of the optimization			
	a) Initialization	b) Evaluation	c) Selection	d) Crossover
160.	_____ Individuals are selected from the population based on their fitness. The more fit individuals have a higher chance of being selected .			
	a) Initialization	b) Evaluation	c) Selection	d) Crossover
161.	Pairs of selected individuals (parents) <i>exchange genetic information</i> to create new individuals (offspring) in the _____ phase of the genetic algorithm			
	a) Mutation	b) Replacement	c) Termination	d) Crossover
162.	_____ Random changes are applied to some of the individuals in the population. This introduces new genetic material and helps explore the solution space more broadly			
	a) Mutation	b) Replacement	c) Termination	d) Crossover
163.	_____ the new offspring and possibly some of the existing individuals are used to form the next generation. The less fit individuals may be replaced by the fitter offspring			
	a) Mutation	b) Replacement	c) Termination	d) Crossover
164.	_____ the algorithm continues to iterate through the selection, crossover, mutation, and replacement steps for a fixed number of generations or until a termination condition is met.			
	a) Mutation	b) Replacement	c) Termination	d) Crossover
165.	_____ is a widely applied concept to deal with uncertainty .			
	a) Fuzzy theory	b) Fuzzy logic	c) Both A&B	d) Fuzzy Sets



166.	_____ concepts that belong to the field of fuzzy set theory , a mathematical framework for dealing with uncertainty and vagueness.			
	a) Fuzzy theory	b) Fuzzy logic	c) Both A&B	d) Fuzzy Sets
167.	_____ theory allows elements to partially belong to a set .			
	a) Fuzzy theory	b) Fuzzy logic	c) Both A&B	d) Fuzzy Sets
168.	_____ Repeat the process for a certain number of generations or until a termination condition is met			
	a) Mutation	b) Replacement	c) Termination	d) Crossover
169.	_____ Replace some individuals in the old population with the new offspring. For simplicity, replace the two least fit individuals.			
	a) Mutation	b) Replacement	c) Termination	d) Crossover
170.	_____ Select individuals based on their fitness. Higher-performing feature subsets are more likely to be selected.			
	a) Selection	b) Replacement	c) Termination	d) Crossover
171.	The difference between the actual output (For YT) and the desired output (Z) for a given set of inputs is an error called _____			
	a) Delta	b) Sigma	c) Alpha	d) None
172.	The _____ function computes the weighted sums of all the input elements entering each processing element.			
	a) Summation	b) Transfer	c) Connection Weights	d) None
173.	_____ is a broad concept. Its objective is to assess an ES's overall value. In addition to assessing acceptable performance levels, it analyzes whether the system would be usable, efficient, and cost-effective			
	a) Evaluation	b) Validation	c) Verification	d) All
174.	_____ is the part of evaluation that deals with the <u>performance of the system</u> , simply stated, it is building the <u>right system</u>			
	a) Evaluation	b) Validation	c) Verification	d) All
175.	_____ is building the <u>system right</u> or substantiating that the system is correctly implemented to its specifications.			
	a) Evaluation	b) Validation	c) Verification	d) All
176.	_____ declarative rules, state all the facts and relationships about a problem.			
	a) Knowledge rules	b) Inference rules	c) Expect rules	d) None
177.	_____ procedural rules, offer advice on how to solve a problem, given that certain facts are known.			
	a) Knowledge rules	b) Inference rules	c) Expect rules	d) None
178.	_____ is a goal-driven approach in which you start from an expectation of what is going to happen and then seek evidence that supports your expectation. Often, this entails formulating and testing intermediate hypotheses (or sub-hypotheses)			
	a) Forward chaining	b) Backward chaining	c) Reinforcement	d) Supervised DS
179.	_____ is a data-driven approach. We start from available information as it becomes available or from a basic idea, and then we try to draw conclusions.			
	a) Forward chaining	b) Backward chaining	c) Reinforcement	d) Supervised DS



180.	_____ is the process of using the rules in the knowledge base along with the known facts to draw conclusions .			
	a) inferencing	b) reasoning	c) Both A&B	d) None
181.	_____ is represented in the form of condition/action pairs: IF this condition (or premise or antecedent) occurs, THEN some action (or result or conclusion or consequence) will (or should) occur.			
	a) Knowledge	b) inferencing	c) reasoning	d) Production
182.	_____ involves the acquisition of knowledge from human experts, books, documents, sensors, or computer files.			
	a) Knowledge Acquisition	b) Knowledge Validation	c) Knowledge Representation	d) Interfacing
183.	_____ Knowledge acquired from experts or induced from a set of data must be represented in a format that is both understandable by humans and executable on computers.			
	a) Knowledge Acquisition	b) Knowledge Validation	c) Knowledge Representation	d) Interfacing
184.	_____ is the process of using the rules in the knowledge base along with the known facts to draw conclusions .			
	a) Inferencing (reasoning)	b) Validation	c) Evaluation	d) None
185.	_____ is where the information is stored in the expert system in form of facts and rules. Basically, a series of IF statements. Here the programmer writes the code for the expert system.			
	a) knowledge base	b) user interface	c) inference engine	d) None
186.	_____ is where the user interacts with the expert system. This is where questions are asked and advised is produced. It can also output justification features of the expert system			
	a) knowledge base	b) user interface	c) inference engine	d) None
187.	_____ <u>applies the facts to the rules</u> and determines the questions to be asked of the user in the user interface and which order to ask them. This is an invisible part of an ESS when the program is run			
	a) knowledge base	b) user interface	c) inference engine	d) None
188.	During the _____ phase, you conduct a thorough analysis of the current decision-making processes within the organization.			
	a) diagnosis	b) Initialization	c) Make/Buy	d) Feasibility study
189.	_____ is structured approach for evaluating and assessing the effectiveness, efficiency, and security of a Decision Support System within an organization.			
	a) DSS audit plan	b) DSS diagnosis	c) DSS planning	d) DSS evaluation
190.	_____ is a rule-based system that provides a solution, usually in one functional area (e.g., finance, manufacturing), to a specific repetitive managerial problem, usually in one industry.			
	a) ADS	b) ES	c) DSS	d) KBS
191.	_____ developed at Stanford University in the early 1980s for medical diagnosis , is the most well-known ES application			
	a) SONAR	b) LMOS	c) XCON	d) MYCIN
192.	In the influence diagram, _____ represented by hexagons and represent the objectives of the decision-maker.			
	a) Decision nodes	b) Chance nodes	c) Value nodes	d) output nodes



193.	In influence diagram, _____ are represented by circles and represent the uncertainties in the problem.			
	a) Decision nodes	b) Chance nodes	c) Value nodes	d) output nodes
194.	In influence diagram, _____ are represented by squares and represent the decisions that need to be made			
	a) Decision nodes	b) Chance nodes	c) Value nodes	d) output nodes
195.	In influence diagram, _____ represent the relationships between the different elements of the problem.			
	a) Decision nodes	b) Chance nodes	c) Value nodes	d) Arcs
196.	_____ is the identification of problems or opportunities for improvement in current decision behavior.			
	a) diagnosis	b) Initialization	c) Make/Buy	d) Feasibility study
197.	_____ is an area of working memory set aside as a database for <u>description of the current problem</u> , as characterized by the input data			
	a) User interface	b) Knowledge base	c) Blackboard	d) Database
198.	The collection of intensive activities encompassing the acquisition of knowledge from human experts and conversion of this knowledge into a repository (commonly called a knowledge base) are called knowledge _____			
	a) acquisition	b) engineering	c) evaluation	d) validation
199.	Random changes are applied to some of the individuals in the population in _____ phase of genetic algorithm.			
	a) Mutation	b) Replacement	c) Termination	d) Crossover
200.	A _____ DSS utilizes a model to solve problems or aid in decision making.			
	a) Model-driven	b) Communication-driven	c) Data-driven	d) Web-based
201.	A _____ are costs that remain the same regardless of the level of production, such as rent, salaries, and insurance.			
	a) Fixed costs	b) Variable costs	c) Selling price	d) None
202.	A _____ are costs that change with the level of production, such as raw materials and direct labor.			
	a) Fixed costs	b) Variable costs	c) Selling price	d) None
203.	The _____ is the price at which the product or service is sold.			
	a) Fixed costs	b) Variable costs	c) Selling price	d) None
204.	_____ analysis can help you to identify trends and patterns in your financial data.			
	a) Ratio	b) Break-even	c) Decision	d) None
205.	_____ analysis can help you to compare your company's performance to that of other companies in your industry.			
	a) Ratio	b) Break-even	c) Decision	d) None
206.	_____ analysis can help you to identify and evaluate alternatives with their respective pros and cons .			
	a) Ratio	b) Break-even	c) Decision	d) None
207.	_____ analysis can help you to evaluate all the alternatives and pick the one that they think is the best.			
	a) Ratio	b) Break-even	c) Decision	d) None



208.	The AHP uses _____ criteria in evaluation.						
	a)	Qualitative	b)	Quantitative	c)	A & B	d) None
209.	_____ relies on pairwise comparisons and the determination of relative importance among criteria and alternatives.						
	a)	AHP	b)	RATIO	c)	MAUA	d) None
210.	A _____ uses a tree-like flowchart of decisions , drawn from left to right, with further branches explaining their consequences, cost involved, event outcomes and utility.						
	a)	decision tree	b)	decision chart	c)	DSS	d) None
211.	In decision tree, _____ node is represented by a square .						
	a)	Choice	b)	Chance	c)	End node	d) None
212.	In decision tree, _____ node is represented by a circle .						
	a)	Choice	b)	Chance	c)	End node	d) None
213.	In decision tree, _____ node is represented by a triangle .						
	a)	Choice	b)	Chance	c)	End node	d) None
214.	A _____ the prediction will be used anytime soon mainly in deterministic models.						
	a)	Short run forecasts	b)	Long run forecasts	c)	Forecasting	d) None
215.	A _____ the prediction is used for long term investment/planning decisions.						
	a)	Short run forecasts	b)	Long run forecasts	c)	Forecasting	d) None
216.	Its _____ a diagrammatic representation of a decision situation, to express the <i>precise nature of relationships</i> between variables and the influence one variable has on others.						
	a)	Influence diagram	b)	Tree diagram	c)	Multi-attribute	d) None
217.	A _____ conduct experiments to identify conditions or situations that approximate the actual conditions.						
	a)	simulation models	b)	Network and optimization models	c)	Mathematical models	d) None
218.	A _____ are integrated into a DSS when decisions regarding resource allocation , project control, location, scheduling, etc.						
	a)	simulation models	b)	Network and optimization models	c)	Mathematical models	d) None
219.	_____ analysis takes a single snapshot of a situation and assumes that it will remain stable all through and won't change.						
	a)	Static	b)	Dynamic	c)	A & B	d) None
220.	. _____ analysis method considers that the situation changes over time , due to any reason, such as cost, rules and regulations, time, etc.						
	a)	Static	b)	Dynamic	c)	A & B	d) None
221.	A _____ DSS is a computer-based reasoning system						
	a)	Knowledge-Driven	b)	Model-driven	c)	Communication-driven	d) Building-driven
222.	. _____ DSS provides information, comprehension and suggestions to users						
	a)	Knowledge-Driven	b)	Model-driven	c)	Communication-driven	d) Building-driven



223.	A _____ computer system that imitates the decision-making capability of a human expert .				
	a)	Expert System	b)	Knowledge Discovery	c) Data Mining d) None
224.	A _____ data mining model predicts which prospects are likeliest to respond to a particular stimulus.				
	a)	Predictive	b)	Descriptive	c) Mathematical d) None
225.	A _____ model describes patterns in existing data to create significant demographics subgroups, which can then be used for target marketing.				
	a)	Predictive	b)	Descriptive	c) Mathematical d) None
226.	A _____ is the data about the data .				
	a)	data dictionary (metadata)	b)	data base	c) data type d) None
227.	A _____ is the self-describing nature of the database that provides program-data independence.				
	a)	System Catalog	b)	System Analysis	c) System dictionary d) None
228.	_____ data dictionary (also called integrated data dictionary) is managed automatically by the database management software.				
	a)	Active	b)	Passive	c) A & B d) None
229.	A _____ data dictionary (also called non-integrated data dictionary) is the one used only for documentation purposes.				
	a)	Active	b)	Passive	c) A & B d) None
230.	A _____ is an organized approach to design specialized decision support systems, more precisely their user interfaces.				
	a)	ROMC	b)	AHP	c) RATIO d) MAUA
231.	A _____ is systematic move towards designing the representation, operation, memory and control mechanisms of a large decision support system				
	a)	ROMC	b)	AHP	c) RATIO d) MAUA
232.	_____ is about presenting information or results for that matter, in a structured way.				
	a)	Representation	b)	Operation	c) Memory Aid d) Control Aid
233.	_____ stage in user interface design focuses on specific tasks performed by/with a decision support system.				
	a)	Representation	b)	Operation	c) Memory Aid d) Control Aid
234.	_____ is the DSS mechanism that must give users a link to data warehouse that aids their memory				
	a)	Representation	b)	Operation	c) Memory Aid d) Control Aid
235.	_____ is provided to users, so that they can effectively use <u>representations, operations and memory</u> aids.				
	a)	Representation	b)	Operation	c) Memory Aid d) Control Aid
236.	The _____ phase involves defining the purpose and objectives of the DSS.				
	a)	initialization	b)	Diagnosis	c) DSS Audit plan d) None
237.	A _____ identify the specific decision-making problems or opportunities that the DSS will address				
	a)	initialization	b)	Diagnosis	c) DSS Audit plan d) None



238.	A _____ specifies incremental development with constant feedback from potential users.			
	a) Rapid Application Development (RAD)	b) SDLC	c) End-User Development	d) Control Aid
239.	Creating a _____ for a DSS design involves quickly building a simplified version of the system to demonstrate its key features and functionality.			
	a) SDLC	b) Rapid Prototype	c) End-User Development	d) Make Or Buy
240.	A _____ emphasizes active involvement of end-users in the system development process.			
	a) SDLC	b) Rapid Prototype	c) End-User Development	d) Make Or Buy
241.	In the _____ step, you decide whether to build the DSS in-house or purchase an existing DSS solution from a vendor.			
	a) SDLC	b) Rapid Prototype	c) End-User Development	d) Make Or Buy
242.	A _____ managers with unstructured and semi structured problems by focusing on the information needs of <u>senior management</u> .			
	a) ESS	b) MIS	c) TPS	d) DSS
243.	_____ information is stored in the expert system inform of facts and rules.			
	a) knowledge base	b) user interface	c) inference engine	d) None
244.	A _____ is the collection and storage of <i>structured</i> (facts, rules, regulations, characteristics, functions, procedures, and relationships) and <i>unstructured</i> information that will be used by a DSS in decision making.			
	a) knowledge base	b) Knowledge Acquisition	c) Data Mining	d) Knowledge Engineer
245.	_____ user interacts with the expert system.			
	a) knowledge base	b) user interface	c) inference engine	d) None
246.	_____ applies the facts to the rules and determines the questions to be asked of the user in the user interface and which order to ask them.			
	a) knowledge base	b) user interface	c) inference engine	d) None
247.	A _____ is a software system to simplify the conception and development of application interfaces between application systems.			
	a) knowledge base	b) user interface	c) interface engine	d) None
248.	A _____ is an approach to discovery and problem solving by employing <i>practical methods</i>			
	a) knowledge base	b) user interface	c) inference engine	d) Heuristic
249.	A _____ sheds light on a project's operational, economical, technological, scheduling and legal viability .			
	a) Feasibility Analysis	b) System Development	c) UI Development	d) None
250.	In _____ As a DSS designer, you must try reducing the execution time for a command given and action performed. Maximize the pace of execution to minimize the wastage of time.			
	a) Execution Time	b) Versatility	c) Adaptability	d) Learning Time



251.	A _____ mean that a decision support system must be resourceful enough to perform the entire range of tasks that a decision maker needs to perform when making a decision using DSS.				
	a)	Execution Time	b)	Versatility	c) Adaptability d) Learning Time
252.	A _____ decision support system should be smart enough to adapt according to the most prominent habits of its user.				
	a)	Execution Time	b)	Versatility	c) Adaptability d) Learning Time
253.	A _____ DSS user interface should be simple enough to reduce the learning time of its users, so that they can use it to its full capacity as soon as possible.				
	a)	Execution Time	b)	Versatility	c) Adaptability d) Learning Time
254.	A _____ DSS user interface must have a uniform theme throughout. It should <u>offer the same look</u> and feel and command throughout .				
	a)	Execution Time	b)	Versatility	c) Adaptability d) Uniformity of Command
255.	A _____ DSS built by you, he or she expects complete on and off line support from you. A DSS depends upon the quality of support offered.				
	a)	Execution Time	b)	Versatility	c) Uniformity of Command d) Quality of Help
256.	A person has limitations when it comes to remembering numbers . <u>A good UI design takes the memory load off the user mind.</u>				
	a)	Memory Load	b)	Ease of Recall	c) Adaptability d) Quality of Help
257.	A _____ user comes back to DSS after long, it must help him/her recall what was done previously. It helps them achieve the same pace in a shortened time.				
	a)	Memory Load	b)	Ease of Recall	c) Adaptability d) Quality of Help
258.	_____ Keep things simple and keep the commands visual so that the user <u>doesn't need to remember anything</u> .				
	a)	Memory Load	b)	Ease of Recall	c) Fatigue d) Quality of Help
259.	A _____ Anticipate errors that a user may perform when using a decision support system. <u>Provide them the control to reverse the action and help to guide them what to do next.</u>				
	a)	Memory Load	b)	Ease of Recall	c) Fatigue d) Errors
260.	A _____ are interrelated terms used for the process of extracting valuable knowledge and discovering patterns , in order to transform the knowledge into easily comprehensible structure for further use.				
	a)	Knowledge Discovery	b)	Data Mining	c) Expert System d) Development Environment
261.	A _____ is a process of collection, storing and analysis of data and <u>not finding patterns</u> .				
	a)	Knowledge Discovery	b)	Data Mining	c) Expert System d) Development Environment
262.	A _____ the environment in which a decision support system is developed				
	a)	Knowledge Discovery	b)	Data Mining	c) Expert System d) Development Environment
263.	A _____ is a subject matter expert who has expertise/authority in a particular domain .				
	a)	Domain Expert	b)	Data Mining	c) Expert System d) Development Environment

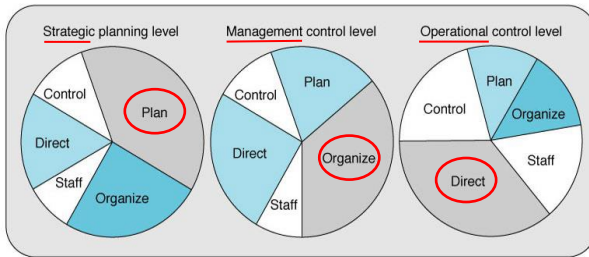


264.	A _____ is technical expert who integrates knowledge into a computer system when developing a decision support system, to solve complex problems that require human expertise.							
	a)	Domain Expert	b)	Data Mining	c)	Development Environment	d)	Knowledge Engineer
265.	A _____ is extraction/mining of knowledge from various sources, such as experts, databases, and external programs.							
	a)	Domain Expert	b)	Knowledge Acquisition	c)	Development Environment	d)	Knowledge Engineer
266.	_____ contain rules about rules and thus are also <u>called meta-rules</u> .							
	a)	Inference rules	b)	Evaluation	c)	Knowledge rules	d)	None
267.	The ES analyzes the problem by looking for the <u>facts that match</u> the _____ part of its IF-THEN rules.							
	a)	IF	b)	THEN	c)	What-If	d)	Sensitivity
268.	A system that integrates knowledge from experts is commonly _____							
	a)	knowledge-based Decision Support System (KBDSS)	b)	Intelligent Decision Support System (IDSS)	c)	A & B	d)	Decision Support System (DSS)
269.	The building blocks of ADS are _____.							
	a)	Inference rules	b)	Business rules	c)	DSS	d)	Decision
270.	_____ collection of concepts and ideas that are related to the development of intelligent systems							
	a)	AI	b)	DB	c)	ES	d)	None
271.	A _____ computer-based information systems that use expert knowledge to attain high-level decision performance in a narrowly defined problem domain .							
	a)	EIS	b)	DB	c)	ES	d)	None
272.	A _____ is the extensive, <i>task-specific</i> knowledge that experts possess .							
	a)	expert	b)	Expertise	c)	Implicit	d)	None
273.	_____ the accumulation , transfer, and transformation of problem-solving expertise from experts or documented knowledge sources to a computer program for constructing or expanding the knowledge base.							
	a)	knowledge base	b)	Knowledge acquisition	c)	blackboard	d)	None
274.	According to the _____, knowledge engineering deals with the steps necessary to build expert systems (i.e., knowledge acquisition, knowledge representation, knowledge validation, inferencing, and explanation/justification).							
	a)	narrow perspective	b)	broad perspective	c)	metaknowledge	d)	None
275.	According to the _____, the term describes the entire process of developing and maintaining any intelligent systems							
	a)	narrow perspective	b)	broad perspective	c)	metaknowledge	d)	None
276.	_____ mean information about how experts use their knowledge to solve problems and about problem-solving procedures in general.							
	a)	Knowledge Base	b)	Information	c)	Metaknowledge	d)	Metadata
277.	The <u>perceptron model fell into disfavor</u> because it was unable to solve the simple _____ problem.							
	a)	AND	b)	OR	c)	NAND	d)	XOR



278.	_____ rules are the most popular form of knowledge representation for expert systems.			
	a) Production	b) Knowledge	c) Development	d) Inference
279.	_____ is the knowledge that has been codified (i.e., documented) in a form that can be distributed to others or transformed into a process or strategy without requiring interpersonal interaction.			
	a) Explicit	b) Tacit	c) Procedural	d) Declarative
280.	_____ is usually either localized within the brain of an individual or embedded in the group interactions within a department or a branch office.			
	a) Explicit	b) Tacit	c) Procedural	d) Declarative
281.	An ES builder uses the _____ to build the necessary components of the ES and to populate the knowledge base with appropriate representation of the expert knowledge.			
	a) development environment	b) consultation environment	c) Evaluation	d) explanation subsystem
282.	A nonexpert uses the _____ to obtain advice and to solve problems using the expert knowledge embedded into the system			
	a) development environment	b) consultation environment	c) Evaluation	d) explanation subsystem
283.	An _____ DSS <i>provides stakeholders with access to a company's intranet</i> and authority or privileges to use specific DSS capabilities.			
	a) Inter-Organizational	b) Intra-Organizational	c) Internet	d) Intranet
284.	_____ are designed for use by individuals in a company as "stand-alone DSS" or for use by a group of managers in a company as a Group or Enterprise-Wide DSS.			
	a) Inter-Organizational	b) Intra-Organizational	c) Internet	d) Intranet

True/False:

1.	<i>Management <u>reporting</u> systems <u>report</u> on the past and the present, rather than projecting the future.</i>	T
2.	<i>Executive information systems support <u>top managers</u> with conveniently displayed summarized information, customized for them.</i>	T
3.	The core management activates for planning in the <u>management level</u> of organization (organizing)	F
		
4.	<i><u>Decision</u> is a choice made between alternative courses of action in a situation of uncertainty.</i>	T
5.	<i>Decision is a <u>cognitive process</u> that results in the selection of a course of action among several alternative scenarios (Decision-making)</i>	F
6.	Pareto chart helps you to identify all possible causes of the problem. (Cause-and-effect)	F
7.	Forward Analysis Sensitivity Models known as <u>goal</u> seeking analysis. (Backward)	F
8.	Decision-making under uncertainty is more difficult because of insufficient information	T



9.	The state of nature are the <u>different possible strategies</u> the decision maker can employ. (<i>decision alternatives</i>)	F
10.	States of nature could be defined as low demand and high demand.	T
11.	Alternatives could decide to build as small, medium, or large condominium complex.	T
12.	Payoffs are the profit for each alternative under each potential state of nature is going to be determined.	T
13.	A model driven DSS may employ a single model or a combination of two or more models.	T
14.	Model-driven DSS are generally data intensive . (<i>not data intensive</i>)	F
15.	AHP is suitable for qualitative and quantitative information	T
16.	In a Decision Support System, the primary focus is often on the computerized components of the system	T
17.	Information systems capture data from the <i>organization (internal data)</i> and its <i>environment (external data)</i> .	T
18.	Most of the data captured by information systems relates to the operations of the organization itself	T
19.	The <i>fewer resources</i> a system consumes in producing given outputs, the more effective it is. (<i>efficient</i>)	F
20.	The objective of management reporting systems is to provide routine information to managers	T
21.	The decision support system is <u>under user control</u> , from early beginning to final putting into use and daily use	T
22.	DSS refers to a class of systems, which supports the process of decision-making and does not always give a decision itself	T
23.	To increase the production level by 40 percent using the backward sensitivity analysis, first, the target value for the production level can be set	T
24.	An Inter-Organizational DSS provides stakeholders with access to a company's intranet and authority or privileges to use specific DSS capabilities	T
25.	ERP is <u>NOT</u> a Decision Support System	T
26.	In the payoff table, the rows correspond to the possible decision alternatives	T
27.	In pro forma financial model Costs are estimated based on past data, gross sales are predicted, and profit or loss is then calculated on these relationships	T
28.	DSS with simulation models conduct experiments to identify conditions or situations that approximate the actual conditions	T
29.	The passive data dictionary (also called integrated data dictionary) is the one used only for documentation purposes. (<i>non-integrated</i>)	F



FINAL QUESTIONS (AHP)

1.	In AHP, If the CR is greater than the judgments are untrustworthy because they are too close for comfort to randomness and the exercise is valueless or must be repeated.							
	a)	0.1	b)	0.01	c)	50%	d)	0.2
2.	According to figure 1, the decision maker will recommend buyingbased on AHP method.							
	a)	Escort	b)	Miata	c)	Civic	d)	Saturn
3.	According to the choice in question 2, what is the weight of the selected model?							
	a)	0.30	b)	0.33	c)	0.35	d)	0.39

AHP Last step Example:

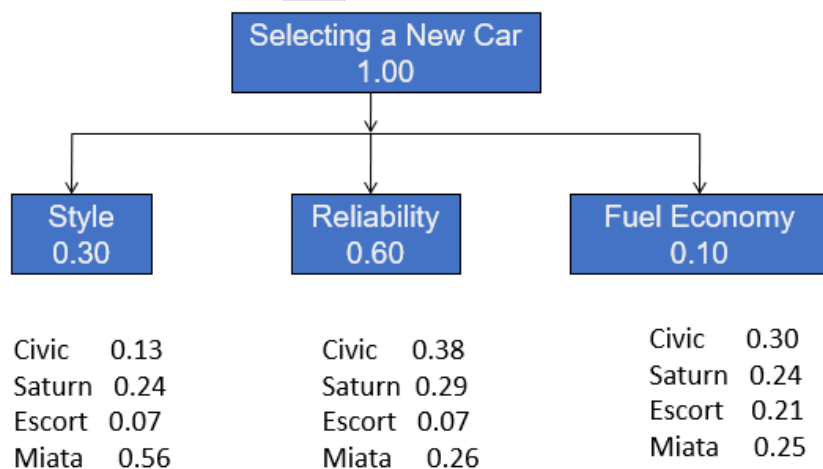


Figure 1

In this figure, what will the decision maker recommend buying?

العربية: decision alternatives التي موجوده عندنا اسمها بدائل القرار civic, Saturn, escort, Miata
 وكل معيار زي ال style, reliability, fuel economy دول ليهم ارقام والأرقام دي اسمها criteria weights
 هناخد القيم بتاعت الصفوف ونعملها مصفوفة matrix ونضربها في عمود ال criteria weights ونختار اكبر قيمة في الناتج.

$$\begin{matrix} \text{Civic} \\ \text{Saturn} \\ \text{Escort} \\ \text{Miata} \end{matrix} \begin{bmatrix} 0.13 & 0.38 & 0.30 \\ 0.24 & 0.29 & 0.24 \\ 0.07 & 0.07 & 0.21 \\ 0.56 & 0.26 & 0.25 \end{bmatrix} \times \begin{bmatrix} 0.30 \\ 0.60 \\ 0.10 \end{bmatrix} = \begin{bmatrix} 0.30 \\ 0.27 \\ 0.08 \\ \mathbf{0.35} \end{bmatrix}$$

The decision maker will recommend buying **Miata**

The weight of the selected model is **0.35**



FINAL QUESTIONS

(Cost-Profit)

Table 1: Profit payoff

		States of nature			
		S1	S2	S3	S4
Alternatives	A1	10	9	14	25
	A2	3	6	2	- 4
	A3	5	4	1	17
	A4	4	2	19	7

1.	According to table 1, the decision maker will choose, based on optimistic approach.							
	a)	A1	b)	A2	c)	A3	d)	A4

Optimistic : max → max

هنختار هنا اكبر القيم في A, B, C 🚨

A1: Max = 25.
A2: Max = 6.
A3: Max = 17.
A4: Max = 19.

Decision: A1

و هنختار اكبر قيمه فيهم الثلاثه 🚨

2.	According to table 1, the decision maker will choose, based on pessimistic approach.							
	a)	A1	b)	A2	c)	A3	d)	A4

Pessimistic : min → max

هنختار هنا اصغر القيم في A, B, C 🚨

A1: Min = 9.
A2: Min = -4.
A3: Min = 1.
A4: Min = 2.

Decision: A1

و هنختار اكبر قيمه فيهم الثلاثه 🚨

3.	According to table 1, the decision maker will choose, based on regret approach.							
	a)	A1	b)	A2	c)	A3	d)	A4

Regret : max → max → min

أكبر قيمه في كل عمود 🚩

S1: Best payoff = 10

S2: Best payoff = 9

S3: Best payoff = 19

S4: Best payoff = 25

Regret = Best payoff in state - Payoff of the alternative (من الجدول)

Alternative	S1	S2	S3	S4
A1	10-10= 0	9-9= 0	19-14= 5	25-25= 0
A2	10-3= 7	9-6= 3	19-2= 17	25-(-4) = 29
A3	10-5= 5	9-4= 5	19-1= 18	25-17 = 8
A4	10-4 =6	9-2 = 7	19 - 19 = 0	25-8 = 18

أكبر قيمه في كل صف 🚩

A1: Max regret = 5

A2: Max regret = 29

A3: Max regret = 18

A4: Max regret = 18

Decision: A1.

4.	According to table 1, the decision maker will choose, based on expected value approach.							
	a)	A1	b)	A2	c)	A3	d)	A4
5.	According to table 1, the decision maker will achieve, based on expected value approach.							
	a)	1.75	b)	6.75	c)	8	d)	14.5

Expected value approach:

Alternative	S1	S2	S3	S4
A1	10	9	14	25
A2	3	6	2	-4
A3	5	4	1	17
A4	4	2	19	8
Probabilities	P(S1)=0.25	P(S2)=0.25	P(S3)=0.25	P(S4)=0.25

EV A1 = (10×0.25)+(9×0.25)+(14×0.25)+(25×0.25) = 14.5

EV A2 = (3×0.25)+(6×0.25)+(2×0.25)+(-4×0.25) = 1.75

EV A3 = (5×0.25)+(4×0.25)+(1×0.25)+(17×0.25) = 6.75

EV A4 = (4×0.25)+(2×0.25)+(19×0.25)+(7×0.25) = 8

Decision: A1.



6.	According to table 1, the EVPI equals _____.							
	a)	9	b)	14	c)	1.25	d)	7.75

EVPI approach:

Table 1: Profit payoff

		States of nature			
		S1	S2	S3	S4
Alternatives	A1	10	9	14	25
	A2	3	6	2	- 4
	A3	5	4	1	17
	A4	4	2	19	7

🚨 هنختار اكبر رقم في كل عمود state و هنضربه في الاحتمال بتاعه

$$EVwPI = 0.25(10) + 0.25(9) + 0.25(19) + 0.25(25) = 15.75$$

الخطوة دي معتمده على اللي قبلها اللي طلعتا منها ال expected value

🚨 عشان السؤال profit هنا بنطرح اكبر ev

1. Subtract **highest EV** from EVwPI to calculate EVPI:

$$EVPI = | EVwPI - Highest EV | = 15.75 - 14.5 = 1.25$$

FINAL QUESTIONS (Perceptron)

(الأزرق: معطيات السؤال، الأحمر: شرط الـ Y)

1.	According to table 2, The threshold value (0.5) sets the output Y to 1 in the next row if the weighted sum of inputs is greater than 0.5; otherwise, Y is set to 0, Alpha =0.2, initially w1=0.1, w2=0.3, so the new weights after the first step are			
	a) w1=0.1, w2=0.3	b) w1=0.1, w2=0.5	c) w1=0.3, w2=0.3	d) w1=0.3, w2=0.5
2.	According to table 2, The threshold value (0.5) sets the output Y to 1 in the next row if the weighted sum of inputs is greater than 0.5; otherwise, Y is set to 0, Alpha =0.2, initially w1=0.1, w2=0.3, so the new weights after the second step are			
	a) w1=0.1, w2=0.3	b) w1=0.1, w2=0.5	c) w1=0.5, w2=0.7	d) w1=0.3, w2=0.5

Inputs			Z
Case	X ₁	X ₂	Desired Results
1	0	0	0
2	0	1	1 (positive)
3	1	0	1 (positive)
4	1	1	1 (positive)

دي المسألة اللي هنحلها

الغرض من السؤال ان عندي neural network بعلمها انها تطلع لي النواتج المرجوة Z في خانة النواتج الحقيقية Y.

معطيات السؤال Given

Inputs → X₁, X₂

Initial Weights → W₁ = 0.1, W₂ = 0.3

Desired Results/Outputs → Z

Threshold → 0.5

Alpha → 0.2

البيصجين



خطوات الحل:

مبدأياً كذا ال STEP جواها 4 cases وال case يعني سطر
لما بتخلص كل ال cases يبقى انت خلصت step
بس لو ال Y مبتساويزش ال Z في ال step كلها يبقى بتعمل step جديد

بس في الامتحان لو طلب منك weights of the first step بتعمل ال first step بس
وتجيب ال final weight بتاع اخر case

أما لو طلب ال weights of the second step يبقى بتعمل ال first step وال second step
وتجيب ال final weight بتاع اخر case في ال second step

حل أول سطر

$$(0.1*0) + (0.3*0) < 0.5 \implies y=0$$

$$0-0=0 \implies \text{Delta}$$

$$w_1(\text{final}) = 0.1 + 0.2 * 0 * 0 = 0.1$$

$$w_2(\text{final}) = 0.3 + 0.2 * 0 * 0 = 0.3$$

كرر الخطوات !!

Z بتختلف لو المسألة كانت or او and
هي هنا or

1. بنطلع حاجه اسمها weighted sum

$$X_1 * W_1 + X_2 * W_2$$

لو ال weighted sum اكبر من 0.5 يبقى Y=1
غير كذا يبقى Y=0

2. بعد ما طلعتنا ال Y بنطلع ال $\text{delta} = Z - Y$
وال Z مكتوبه جاهزه

3. بعد ما طلعتنا ال delta بنطلع ال final weight

$$W_{\text{initial}} + \alpha * \text{delta} * X$$

وبنطلعها ل W1 و W2

اول اثنين جم من راس السؤال
بيكتبوا في اول سطر بس

ثالث خطوة:

ثاني خطوة:

$$W_{\text{initial}} + \alpha * \text{delta} * X$$

Step				Initial		Y	Z - Y Delta	Final	
	X ₁	X ₂	Z	*W ₁	W ₂			W ₁	W ₂
1	0	0	0	0.1	0.3	0	0.0	0.1	0.3
	0	1	1	0.1	0.3	0	1.0	0.1	0.5
	1	0	1	0.1	0.5	0	1.0	0.3	0.5
	1	1	1	0.3	0.5	1	0.0	0.3	0.5
2	0	0	0	0.3	0.5	0	0.0	0.3	0.5
	0	1	1	0.3	0.5	0	1.0	0.3	0.7
	1	0	1	0.3	0.7	0	1.0	0.5	0.7
	1	1	1	0.5	0.7	1	0.0	0.5	0.7

جت من الرسمة وبكرها في كل step

Parameters: Alpha = 0.2, Threshold = 0.5

ال final weights القديم في اول سطر
بيبقوا initial weights في ثاني سطر وهكذا

Weights of
First Step

Weights of
Second Step





WRITTEN EXPLANATION

Cost-profit, AHP

1- Cost-profit

الطلبات:

(اكيد مش هيطلب كل حاجة 😊)

A. Optimistic, Pessimistic, and Regret Approaches

B. Expected Value Approach and EVPI

C. Decision Tree

المطلوب الاول A : عاوز **Optimistic, Pessimistic, and Regret Approaches**

“ Optimistic

“ Conservative (Pessimistic)

“ Regret

	Profit	Cost
Optimistic	Max → max (maximax) (max of max)	Min → min (minimin) (min of min)
Conservative (pessimistic)	Min → max (maximin) (max of min)	Max → min (minimax) (min of max)
Regret	Max → max → min (min of max of max)	Min → max → min (min of max of min)

في البداية المسائل دي نوعين اما **Profit ربح** أو **Cost تكلفة** ويحدد في رأس السؤال

A.1 لاحظ ان المسألة Profit



- A. According to the following profit payoff table, using the optimistic, pessimistic and regret approaches, which solution decision maker choose form A,B, and C
- B. If decision maker know that the probabilities of $s_1=0.2$, $s_2=0.3$, and $s_3=0.5$, which decision according to expected value approach, and the value of EVPI?

	S1	S2	S3
A	4	-4	9
B	5	8	3
C	6	12	16

دي المسألة اللي هنحلها



A.1.1 Optimistic Approach (Maximax) ($\max \rightarrow \max$):

- Identify the **maximum payoff** for each alternative:

هنختار هنا أكبر القيم في A, B, C

A: Payoffs = 4, -4, 9 $\rightarrow$ Max = 9.

B: Payoffs = 5, 8, 3 $\rightarrow$ Max = 8.

C: Payoffs = 6, 12, 16 $\rightarrow$ Max = **16**.

- Choose the alternative with the **maximum of these values**:

وهنختار أكبر قيمة فيهم الثلاثة

- Decision: C.**

A.1.2 Pessimistic Approach (Maximin) ($\min \rightarrow \max$):

- Identify the **minimum payoff** for each alternative:

هنختار هنا أصغر القيم في A, B, C

- A:** Payoffs = 4, -4, 9 $\rightarrow$ Min = -4.

- B:** Payoffs = 5, 8, 3 $\rightarrow$ Min = 3.

- C:** Payoffs = 6, 12, 16 $\rightarrow$ Min = 6.

- Choose the alternative with the **maximum of these minimums**:

وهنختار أكبر قيمة فيهم الثلاثة

- Decision: C.**

A.1.3 Regret Approach (Minimax Regret) (max → max → min):

- 1- هـنختار أكبر رقم في كل state (عمود)
- 2- بعد كذا هـنطرح (الرقم اللي اخترناه من S1 - قيم العمود)
- 3- هـنختار أكبر القيم في كل "صف"
- 4- وفي الآخر هـنختار أقل قيمه فيهم

a. Construct the Regret Table:

1. Determine the **best payoff for each state (S1, S2, S3):**

- **S1:** Best payoff = 6
- **S2:** Best payoff = 12
- **S3:** Best payoff = 16

2. Calculate the regret using the formula:

Regret = Max payoff in state – Payoff of the alternative (من الجدول)

1. Regret Table:

Alternative	S1	S2	S3
A	$6 - 4 = 2$	$12 - (-4) = 16$	$16 - 9 = 7$
B	$6 - 5 = 1$	$12 - 8 = 4$	$16 - 3 = 13$
C	$6 - 6 = 0$	$12 - 12 = 0$	$16 - 16 = 0$

Identify the **maximum regret** for each alternative:

خلي بالك "أكبر قيمه في كل صف"

A: Max regret = 16

B: Max regret = 13

C: Max regret = 0

- ♦ Choose the alternative with the ****minimum** of the maximum regrets**:

- **Decision: C.**

A.2 في حالة ان المسألة كانت Cost:

A.2.1 Optimistic Approach (Minimin) ($\min \rightarrow \min$):

- Identify the **minimum cost** for each alternative:

هنا أصغر القيم في A, B, C

- A: Costs = 4, -4, 9 $\rightarrow$ Min = -4.
- B: Costs = 5, 8, 3 $\rightarrow$ Min = 3.
- C: Costs = 6, 12, 16 $\rightarrow$ Min = 6.

- Choose the alternative with the **minimum of these values**:

هنا أصغر قيمة فيهم الثلاثة

- Decision: A.

A.2.2 Pessimistic Approach (Minimax) ($\max \rightarrow \min$):

- Identify the **maximum cost** for each alternative:

هنا أكبر القيم في A, B, C

- A: Costs = 4, -4, 9 $\rightarrow$ Max = 9.
- B: Costs = 5, 8, 3 $\rightarrow$ Max = 8.
- C: Costs = 6, 12, 16 $\rightarrow$ Max = 16.

- Choose the alternative with the **minimum of these maximums**:

هنا أصغر قيمة فيهم الثلاثة

- Decision: B.

A.2.3 Regret Approach (Minimax Regret) (min → max → min):

هنا بقا : 

- 1- هنتار أقل رقم في كل state (عمود)
- 2- بعد كذا هنتارح (قيم العمود - الرقم اللي اخترناه من S1) 3- هنتار أكبر القيم في كل "صف"
- 4- وفي الآخر هنتار أقل قيمه فيهم

1. Determine the **minimum cost for each state (S1, S2, S3):**

خلي بالك "أصغر قيمة في كل عمود" 

- **S1:** Min cost = 4
- **S2:** Min cost = -4
- **S3:** Min cost = 3

2. Calculate the regret using the formula:

Regret = Payoff of the alternative (من الجدول) – Min payoff in state

Alternative	S1	S2	S3
A	4-4= 0	-4-(-4)= 0	9-3= 6
B	5-4= 1	8-(-4)= 12	3-3= 0
C	6-4= 2	12-(-4)= 16	16-3= 13

Identify the **maximum regret** for each alternative:

- **A:** Max regret = 6.
 - **B:** Max regret = 12.
 - **C:** Max regret = 16.
- خلي بالك "أكبر قيمة في كل صف" 

Choose the alternative with the **minimum of these maximum regrets:**

وهنتار أصغر قيمة فيهم الثلاثة

- **Decision: A.**

المطلوب الثاني: عايز قرار بناء على ال **expected value approach** وعايز قيمة ال **EVPI**

B.1 Expected Value for Each Alternative:

Given probabilities:

$$P(S1) = 0.2$$

$$P(S2) = 0.3$$

$$P(S3) = 0.5$$

هنا في السؤال عايطك احتمالات لكل عمود، لو مش مدبك احتمالات هنقسم 1 على عدد ال states (الأعمدة) وده هيبقى احتمال كل state

Use the formula:

$$\text{Expected Value (EV)} = \sum (\text{Payoff} \times \text{Probability})$$

Alternative	S1	S2	S3
A	4	-4	9
B	5	8	3
C	6	12	16
probabilities	p(S1)= 0.2	P(S2)=0.3	P(S3)=0.5

• A:

$$EV A = (4 \times 0.2) + (-4 \times 0.3) + (9 \times 0.5) = 0.8 - 1.2 + 4.5 = 4.1$$

• B:

$$EV B = (5 \times 0.2) + (8 \times 0.3) + (3 \times 0.5) = 1 + 2.4 + 1.5 = 4.9$$

• C:

$$EV C = (6 \times 0.2) + (12 \times 0.3) + (16 \times 0.5) = 1.2 + 3.6 + 8 = 12.8$$

في حالة ان المسألة Profit هنختار الاكبر

- **Decision: C** (since EV C=12.8 is the **highest**).

في حالة ان المسألة Cost هنختار الاصغر

- **Decision: A** (since EV A=4.1 is the **Lowest**).



B.2 Expected Value of Perfect Information (EVPI):

في حالة الprofit 🚨

- For each state, multiply the **best payoff in state** by its probability:

Alternative	S1	S2	S3
A	4	-4	9
B	5	8	3
C	6	12	16
probabilities	p(S1)= 0.2	P(S2)=0.3	P(S3)=0.5

هنختار اكبر رقم في كل عمود state و هنضربه في الاحتمال بتاعه 🚨

$$EV_{wPI} = 0.2(6) + 0.3(12) + 0.5(16) = 12.8$$

الخطوة دي معتمده على اللي قبلها اللي طلعتا منها الexpected value

وهنا بننقص اكبر EV من الEV_{wPI}

- Subtract **highest EV** from EV_{wPI} to calculate EVPI:

$$EVPI = |EV_{wPI} - \text{Highest EV}| = 12.8 - 12.8 = 0$$

في حالة الcost 🚨

- For each state, multiply the **worst payoff in state** by its probability:

Alternative	S1	S2	S3
A	4	-4	9
B	5	8	3
C	6	12	16
probabilities	p(S1)= 0.2	P(S2)=0.3	P(S3)=0.5

هنختار اصغر رقم في كل عمود state و هنضربه في الاحتمال بتاعه 🚨

$$EV_{wPI} = 0.2(4) + 0.3(-4) + 0.5(3) = 1.1$$

الخطوة دي معتمده على اللي قبلها اللي طلعتا منها الexpected value

وهنا بننقص اصغر EV من الEV_{wPI}

- Subtract **lowest EV** from EV_{wPI} to calculate EVPI:

$$EVPI = |EV_{wPI} - \text{Lowest EV}| = |1.1 - 4.1| = 3$$

لو الناتج طلع سالب شيل السالب

2- AHP



فيديو الشرح (انجليزي)

في البداية لازم تحفظ الارقام دي :

1	Equal importance
3	Moderate importance
5	Strong importance
7	Very strong importance
9	Extreme importance
2, 4, 6, 8	Intermediate values
1/3, 1/5, 1/7, 1/9	Values for inverse comparison

	Price Or Cost	Storage Space	Camera	Looks
Price Or Cost	1	5		
Storage Space	1/5	1		
Camera			1	
Looks				1

Price/Cost is of a **strong importance** than Storage space

في السؤال هنا بيقول Price/Cost is of a strong importance than Storage space

- في الجدول 5 = strong importance

- (ابدأ من العمود "الاخضر") العلاقة بين price وال Storage هي 5

- (بص على "الازرق") هتساوي مقلوب الرقم 1/5

نفس الكلام هنا

	Price Or Cost	Storage Space	Camera	Looks
Price Or Cost	1	5	4	
Storage Space	1/5	1		
Camera	1/4		1	
Looks				1

Price/Cost is of **moderate to strong importance** than camera

- بس لو اخذت بالك ان الجدول مفهوش 4 ???

- في المعطى هنا بيقول moderate to strong Importance و بالتالي هتأخذ رقم بين 3 و 5 (بص على الجدول)



AHP

Pair-wise comparison matrix

	Price or Cost	Storage Space	Camera	Looks
Price or Cost	1	5	4	
Storage Space	$\frac{1}{5}$	1	$\frac{1}{2}$	
Camera	$\frac{1}{4}$	2	1	
Looks				1

How important is Camera with respect to Storage space

Camera is of equal to moderate importance than camera storage

Camera - 2x Value
Storage space - x Value

نفس الكلام هنا هتاخذ رقم بين 1 و 3 وهو 2

Camera مثل Storage*

Equal importance
Moderate importance
Strong importance
Very strong importance

بعد لما تخلص خالص دا شكل الجدول اللي هيبقا معاك :

	Price or Cost	Storage Space	Camera	Looks
Price or Cost	1	5	4	7
Storage Space	$\frac{1}{5}$	1	$\frac{1}{2}$	3
Camera	$\frac{1}{4}$	2	1	3
Looks	$\frac{1}{7}$	$\frac{1}{3}$	$\frac{1}{3}$	1

	Price or Cost	Storage Space	Camera	Looks
Price or Cost	$\frac{1}{1.59}$	$\frac{5}{8.33}$	$\frac{4}{5.83}$	$\frac{7}{14}$
Storage Space	$\frac{0.2}{1.59}$	$\frac{1}{8.33}$	$\frac{0.5}{5.83}$	$\frac{3}{14}$
Camera	$\frac{0.25}{1.59}$	$\frac{2}{8.33}$	$\frac{1}{5.83}$	$\frac{3}{14}$
Looks	$\frac{0.14}{1.59}$	$\frac{0.33}{8.33}$	$\frac{0.33}{5.83}$	$\frac{1}{14}$
Sum	1.59	8.33	5.83	14

كل اللي هتعمله دلوقتي انك هتحويل الكسر لعدد عشري

- وتجمع كل عمود

- ** وهنقسم كل رقم في العمود على مجموع العمود كله

المفروض هنا نقسم على 5.83 في camera مثل 7

Normalised Pair-wise comparison matrix

	Price or Cost	Storage Space	Camera	Looks
Price or Cost	0.6289	0.6002	0.6861	0.5000
Storage Space	0.1258	0.1200	0.0858	0.2143
Camera	0.1572	0.2401	0.1715	0.2143
Looks	0.0898	0.0400	0.0572	0.0714

بعد كل دا هيطلع معاك الجدول دا :



Normalised Pair-wise comparison matrix

	Price or Cost	Storage Space	Camera	Looks	Criteria weights
Price or Cost	0.6289	0.6002	0.6861	0.5000	0.6038
Storage Space	0.1258	0.1200	0.0858	0.2143	0.1365
Camera	0.1572	0.2401	0.1715	0.2143	0.1958
Looks	0.0898	0.0400	0.0572	0.0714	0.0646

$$\frac{0.6289+0.6002+0.6861+0.5000}{4} = 0.6038$$

عاوزين نحسب Criteria weights

$$\text{Criteria weights} = \frac{\text{مجموعهم}}{\text{عددهم}}$$

دلوقتي عاوزين نحسب Consistency

فاكر الجدول الاول؟؟ :

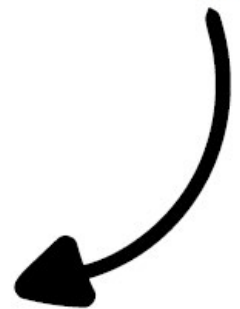
	Price or Cost	Storage Space	Camera	Looks
Price or Cost	1	5	4	7
Storage Space	$\frac{1}{5}$	1	$\frac{1}{2}$	3
Camera	$\frac{1}{4}$	2	1	3
Looks	$\frac{1}{7}$	$\frac{1}{3}$	$\frac{1}{3}$	1

كل عامود هتتضربه في Criteria weights بتاعته

وهتجمع كل صف Weighted Sum Value <=

	Price or Cost	Storage Space	Camera	Looks	Weighted Sum value
Price or Cost	0.6038	0.6825	0.7832	0.4522	2.5217
Storage Space	0.1208	0.1365	0.0979	0.1938	0.5490
Camera	0.1510	0.2730	0.1958	0.1938	0.8136
Looks	0.0863	0.0455	0.0653	0.0646	0.2616

$$0.6038+0.6825+0.7832+0.4522 = 2.5217$$



دلوقتي معانا Consistency & Criteria weights :

	Price or Cost	Storage Space	Camera	Looks	Weighted Sum value	Criteria weights		
Price or Cost	0.6038	0.6825	0.7832	0.4522	2.5217	0.6038	$\frac{2.5217}{0.6038}$	4.1762
Storage Space	0.1208	0.1365	0.0979	0.1938	0.5490	0.1365	$\frac{0.5490}{0.1365}$	4.0225
Camera	0.1510	0.2730	0.1958	0.1938	0.8136	0.1958	$\frac{0.8136}{0.1958}$	4.1553
Looks	0.0863	0.0455	0.0653	0.0646	0.2616	0.0646	$\frac{0.2616}{0.0646}$	4.0488

هنقسم Weighted Sum Value على Criteria weights

$$\lambda_{max} = \frac{4.1762 + 4.0225 + 4.1553 + 4.0488}{4} = 4.1007$$

عشان اجيب λ_{max}

$$\lambda_{max} = \frac{\text{Weighted Sum Value}}{\text{عددهم}}$$

$$\lambda_{max} = 4.1007$$

$$\text{Consistency Index (C.I.)} = \frac{\lambda_{max} - n}{n - 1} = \frac{4.1007 - 4}{4 - 1} = 0.03358$$

where n is the number of compared elements

Consistency Index (C.I.)

St S	n	1	2	3	4	5	6	7	8	9	10
Ca L	RI	0.00	0.00	0.58	0.90	1.12	1.24	1.32	1.41	1.45	1.49

Consistency Ratio

*RI معطى

$$\text{Consistency Index (C.I.)} = 0.03358$$

$$\text{Consistency Ratio} = \frac{\text{Consistency Index (C.I.)}}{\text{Random Index (R.I.)}}$$

$$\text{Consistency Ratio} = \frac{0.03358}{0.90} = 0.037311$$

$$\text{Consistency Ratio (CR)} = 0.037311 < 0.10$$

لو $CR < 0.1$ تبقا consistent

